

Problem Set # 13

S 12.3.8

$$H = \frac{1}{2\mu} \left(\vec{p} - \frac{q}{c} \vec{A}(\vec{x}) \right)^2 \quad (\text{see page 96})$$

$$= \frac{1}{2\mu} \left(p_x + \frac{qB}{2c} y \right)^2 + \frac{1}{2\mu} \left(p_y - \frac{qB}{2c} x \right)^2$$

i. $\vec{B} = \vec{\nabla} \times \vec{A} = \hat{k} (\partial_x A_y - \partial_y A_x)$

$$= \hat{k} \frac{B}{2} (1+1) = B \hat{k}$$

ii. $\dot{x} = \frac{\partial H}{\partial p_x} = \frac{1}{\mu} \left(p_x + \frac{qB}{2c} y \right)$

$$\dot{y} = \frac{\partial H}{\partial p_y} = \frac{1}{\mu} \left(p_y - \frac{qB}{2c} x \right)$$

$$\dot{p}_x = - \frac{\partial H}{\partial x} = - \frac{1}{\mu} \frac{qB}{2c} \left(p_y - \frac{qB}{2c} x \right)$$

$$\dot{p}_y = - \frac{\partial H}{\partial y} = - \frac{1}{\mu} \frac{qB}{2c} \left(p_x + \frac{qB}{2c} y \right)$$

We have

$$\begin{cases} \frac{d}{dt} \left(p_x - \frac{qB}{2c} y \right) = 0 \\ \frac{d}{dt} \left(p_y + \frac{qB}{2c} x \right) = 0 \end{cases}$$

$$\therefore \begin{cases} p_x - \frac{qB}{2c} y = \mu \dot{x} - \frac{qB}{c} y = \text{constant} \\ p_y + \frac{qB}{2c} x = \mu \dot{y} + \frac{qB}{c} x = \text{constant} \end{cases}$$

$$x + iy = z$$

$$\dot{\bar{z}} = \dot{x} + i\dot{y} = \frac{qB}{\mu c} (y - ix) + \text{constant}$$

$$= -i\omega_0 \bar{z} + \text{constant}$$

$$\downarrow$$
$$= -i\omega_0 (\bar{z} - \bar{z}_0)$$

$$\frac{d}{dt} (\bar{z} - \bar{z}_0) = -i\omega_0 (\bar{z} - \bar{z}_0)$$

$$z - z_0 = A e^{-i\omega_0 t} \quad (A = R e^{i\delta})$$

$$x = x_0 + R \cos(\omega_0 t - \delta)$$

$$y = y_0 - R \sin(\omega_0 t - \delta)$$

$$\text{iii} \quad \begin{cases} Q = \frac{c}{qB} P_x + \frac{1}{2} y \\ P = P_y - \frac{qB}{2c} x \end{cases}$$

$$\begin{aligned} [Q, P] &= -\frac{1}{2} [P_x, x] + \frac{1}{2} [y, P_y] \\ &= \hbar i \end{aligned}$$

$$[Q, Q] = [P, P] = 0 \quad (\text{trivially})$$

$$H = \frac{1}{2\mu} \left(P_x + \frac{qB}{2c} y \right)^2 + \frac{1}{2\mu} \left(P_y - \frac{qB}{2c} x \right)^2$$

$$= \frac{1}{2\mu} \left(\frac{qB}{c} \right)^2 Q^2 + \frac{1}{2\mu} P^2$$

$$= \frac{1}{2\mu} P^2 + \frac{1}{2} \mu \omega_0^2 Q^2$$

which is
the
harmonic
oscillator
hamiltonian

Therefore $H |n\rangle = (n + \frac{1}{2}) \hbar \omega_0 |n\rangle$

with $n = 0, 1, 2, \dots$

(Degeneracies may be more complicated than one level for every n)

i.e. $E = (n + \frac{1}{2}) \hbar \omega_0$

IV
$$H = \frac{1}{2\mu} (P_x^2 + P_y^2) + \frac{1}{2\mu} \left(\frac{qB}{2c} \right)^2 (x^2 + y^2)$$

$$+ \frac{qB}{2c\mu} (y P_x - x P_y)$$

$$= \frac{1}{2\mu} (P_x^2 + P_y^2) + \frac{\mu}{2} \left(\frac{\omega_0}{2} \right)^2 (x^2 + y^2)$$

$$- \frac{\omega_0}{2} L_z = H\left(\frac{\omega_0}{2}, \mu\right) - \frac{\omega_0}{2} L_z$$

$$[H, L_z] = 0$$

Cfr. S12.3.7

$$H\left(\frac{\omega_0}{2}, \mu\right) |k, m\rangle = \hbar \frac{\omega_0}{2} (2k + |m| + 1) |k, m\rangle$$

$$L_z |k, m\rangle = m \hbar |k, m\rangle$$

$$k = 0, 1, 2, \dots \quad m = 0, \pm 1, \pm 2, \dots$$

This

$$\begin{aligned} H |k, m\rangle &= \left[\frac{\hbar \omega_0}{2} (2k + |m| + 1) - \frac{\hbar \omega_0 m}{2} \right] |k, m\rangle \\ &= \hbar \omega_0 \left[k + \frac{1}{2} (|m| - m) + \frac{1}{2} \right] |k, m\rangle \end{aligned}$$

Clearly $n = k + \frac{1}{2} (|m| - m)$

S 12.4.4 i. Under an infinitesimal rotation $\vec{\delta\theta}$

$$\vec{V} \rightarrow \vec{V} + \vec{\delta\theta} \times \vec{V} = R(\vec{\delta\theta}) \vec{V}$$

$$V_i \rightarrow V_i + \epsilon_{ijk} \delta\theta_j V_k = R_{ij}(\vec{\delta\theta}) V_j$$

ii.

$$U^\dagger(\vec{\delta\theta}) V_i U(\vec{\delta\theta}) = R_{ij}(\vec{\delta\theta}) V_j$$

$$= \left(1 + \frac{i}{\hbar} \vec{\delta\theta} \cdot \vec{L}\right) V_i \left(1 - \frac{i}{\hbar} \vec{\delta\theta} \cdot \vec{L}\right)$$

$$= V_i + \frac{i}{\hbar} \delta\theta_j [L_j, V_i] + o(\delta\theta^2)$$

$$= V_i + \epsilon_{ijk} \delta\theta_j V_k + o(\delta\theta^2)$$

$$\therefore [L_j, V_i] = i\hbar \epsilon_{jik} V_k$$

$$\text{or } [V_i, L_j] = i\hbar \epsilon_{ijk} V_k$$

S12.5.1

$$\begin{aligned}\psi'_x(x', y') &= \psi_x(x' + \epsilon_z y', y' - \epsilon_z x') \\ &\quad - \epsilon_z \psi_y(x' + \epsilon_z y', y' - \epsilon_z x')\end{aligned}$$

$$\begin{aligned}\psi'_y(x', y') &= \psi_y(x' + \epsilon_z y', y' - \epsilon_z x') \\ &\quad + \epsilon_z \psi_x(x' + \epsilon_z y', y' - \epsilon_z x')\end{aligned}$$

(x', y') can be renamed (x, y)

$$\begin{pmatrix} \psi'_x \\ \psi'_y \end{pmatrix}(x, y) = \begin{pmatrix} 1 & -\epsilon_z \\ \epsilon_z & 1 \end{pmatrix} \begin{pmatrix} \psi_x \\ \psi_y \end{pmatrix}(x + \epsilon_z y, y - \epsilon_z x)$$

$$= \begin{pmatrix} 1 & -\epsilon_z \\ \epsilon_z & 1 \end{pmatrix} \left(1 - \epsilon_z \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \right) \begin{pmatrix} \psi_x \\ \psi_y \end{pmatrix}(x, y)$$

$$\begin{pmatrix} \psi'_x \\ \psi'_y \end{pmatrix} = \begin{pmatrix} \psi_x \\ \psi_y \end{pmatrix} - i\epsilon_z \frac{1}{\hbar} L_z \begin{pmatrix} \psi_x \\ \psi_y \end{pmatrix} - \epsilon_z \begin{pmatrix} 0 & +1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \psi_x \\ \psi_y \end{pmatrix}$$

$$= \left(1 - \frac{i}{\hbar} \epsilon_z J_z \right) \begin{pmatrix} \psi_x \\ \psi_y \end{pmatrix} \quad \text{with}$$

$$J_z = L_z + S_z \quad \text{with} \quad S_z = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

S12.5.3 i. $J_x = \frac{1}{2}(J_+ + J_-)$ $J_y = \frac{1}{2i}(J_+ - J_-)$

$$J_{\pm} |j, m\rangle = \hbar \sqrt{(j \mp m)(j \pm m + 1)} |j, m \pm 1\rangle$$

$$\therefore \langle j, m | J_{\pm} | j, m \rangle = 0$$

$$\therefore \langle j, m | J_x | j, m \rangle = \langle j, m | J_y | j, m \rangle = 0$$

ii. $\langle j, m | (J_x^2 + J_y^2 + J_z^2) | j, m \rangle$

$$= \hbar^2 j(j+1)$$

$$\langle j, m | J_z^2 | j, m \rangle = \hbar^2 m^2$$

$$e^{\frac{i\pi}{\hbar^2} J_z} J_x e^{-\frac{i\pi}{\hbar^2} J_z} = J_y$$

$$e^{\frac{i\pi}{\hbar^2} J_z} J_x^2 e^{-\frac{i\pi}{\hbar^2} J_z} = J_y^2$$

$$\langle j, m | e^{\frac{i\pi}{\hbar^2} J_z} J_x^2 e^{-\frac{i\pi}{\hbar^2} J_z} | j, m \rangle$$

$$= \langle j, m | J_x^2 | j, m \rangle e^{i\frac{\pi}{2}(m-m)}$$

$$= \langle j, m | J_x^2 | j, m \rangle = \langle j, m | J_y^2 | j, m \rangle$$

Hence

$$\begin{aligned}\langle j, m | J_x^2 | j, m \rangle &= \langle j, m | J_y^2 | j, m \rangle \\ &= \frac{1}{2} \hbar^2 [j(j+1) - m^2]\end{aligned}$$

iii. Eq. (9.2.9) says

$$\langle (\Omega - \langle \Omega \rangle)^2 \rangle \langle (\Lambda - \langle \Lambda \rangle)^2 \rangle \geq |\langle (\Omega - \langle \Omega \rangle)(\Lambda - \langle \Lambda \rangle) \rangle|^2$$

$$\begin{aligned}\langle (J_x - \langle J_x \rangle)^2 \rangle &= \langle J_x^2 \rangle - \langle J_x \rangle^2 \\ &= \frac{1}{2} \hbar^2 [j(j+1) - m^2]\end{aligned}$$

Similarly

$$\langle (J_y - \langle J_y \rangle)^2 \rangle = \frac{1}{2} \hbar^2 [j(j+1) - m^2]$$

$$\langle j, m | (J_x - \langle J_x \rangle)(J_y - \langle J_y \rangle) | j, m \rangle$$

$$= \langle j, m | J_x J_y | j, m \rangle$$

$$= \frac{1}{4i} \langle j, m | (J_+ + J_-)(J_+ - J_-) | j, m \rangle$$

$$= \frac{1}{4i} \langle j, m | (J_- J_+ - J_+ J_-) | j, m \rangle$$

$$= \frac{\hbar^2}{4i} \left\{ \sqrt{(j+m+1)(j-m)} \sqrt{(j-m)(j+m+1)} - \sqrt{(j-m+1)(j+m)} \sqrt{(j+m)(j-m+1)} \right\}$$

$$= \frac{\hbar^2}{4i} [(j+m+1)(j-m) - (j+m)(j-m+1)]$$

$$= -\frac{\hbar^2}{2i} m$$

$$|\langle (J_x - \langle J_x \rangle)(J_y - \langle J_y \rangle) \rangle|^2$$

$$= \left| -\frac{\hbar^2}{2i} m \right|^2 = \frac{\hbar^4}{4} m^2 \quad \text{whereas}$$

$$\langle (J_x - \langle J_x \rangle)^2 \rangle \langle (J_y - \langle J_y \rangle)^2 \rangle$$

$$= \left(\frac{\hbar^2}{2} [j(j+1) - m^2] \right)^2$$

The inequality is satisfied since

$$j(j+1) - m^2 = j^2 - m^2 + j \geq j \geq |m|$$

iv. When $m = \pm j$,

$$|\langle (J_x - \langle J_x \rangle)(J_y - \langle J_y \rangle) \rangle|^2 = \frac{\hbar^4}{4} j^2$$

and

$$\begin{aligned} & \langle (J_x - \langle J_x \rangle)^2 \rangle \langle (J_y - \langle J_y \rangle)^2 \rangle \\ &= \left(\frac{\hbar^2}{2} [j(j+1) - j^2] \right)^2 = \frac{\hbar^4}{4} j^2 \end{aligned}$$

512.5.5

$$\frac{1}{\hbar} \vec{J}^{(1/2)} = \frac{1}{2} \hbar \vec{\sigma}$$

$$\vec{\theta} = \vec{n} \theta$$

$$\sigma_i \sigma_j = \delta_{ij} 1 + i \epsilon_{ijk} \sigma_k$$

$$(\vec{n} \cdot \vec{\sigma})^2 = n_i n_j \sigma_i \sigma_j = \vec{n} \cdot \vec{n} = 1$$

$$D^{1/2}(\vec{\theta}) = e^{-i \frac{1}{2} \theta \vec{n} \cdot \vec{\sigma}} = \sum_{p=0}^{\infty} \left(\frac{-i\theta}{2} \right)^p \frac{1}{p!} (\vec{n} \cdot \vec{\sigma})^p$$

$$= \sum_{\substack{p=0 \\ \text{even}}}^{\infty} \left(\frac{-i\theta}{2} \right)^p \frac{1}{p!} + (\vec{n} \cdot \vec{\sigma}) \sum_{\substack{p=1 \\ \text{odd}}}^{\infty} \left(\frac{-i\theta}{2} \right)^p \frac{1}{p!}$$

$$= \cos \theta/2 - i (\vec{n} \cdot \vec{\sigma}) \sin \theta/2$$

$$= \cos \theta/2 \quad 1_{2 \times 2} - i \frac{\theta}{\hbar} \hat{\theta} \cdot \vec{J}^{(1/2)} \sin \theta/2$$

$$\frac{1}{\hbar} \vec{J}_x^{(1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\left(\frac{1}{\hbar} \vec{J}_x^{(1)} \right)^2 = \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$\left(\frac{1}{\hbar} J_x^{(1)}\right)^3 = \frac{1}{2\sqrt{2}} \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix} = \frac{1}{\hbar} J_x^{(1)}$$

$$D^{(1)}(\theta_x) = e^{-\frac{i}{\hbar} \theta_x J_x^{(1)}} = \sum_{p=0}^{\infty} \frac{(-i\theta_x)^p}{p!} \left(\frac{1}{\hbar} J_x^{(1)}\right)^p$$

$$= 1_{3 \times 3} + \frac{1}{\hbar} J_x^{(1)} \sum_{\substack{p=1 \\ \text{odd}}}^{\infty} \frac{(-i\theta_x)^p}{p!}$$

$$+ \left(\frac{1}{\hbar} J_x^{(1)}\right)^2 \sum_{\substack{p=2 \\ \text{even}}}^{\infty} \frac{(-i\theta_x)^p}{p!}$$

$$= 1_{3 \times 3} - \frac{1}{\hbar} J_x^{(1)} \sin \theta_x + \left(\frac{1}{\hbar} J_x^{(1)}\right)^2 (\cos \theta_x - 1)$$